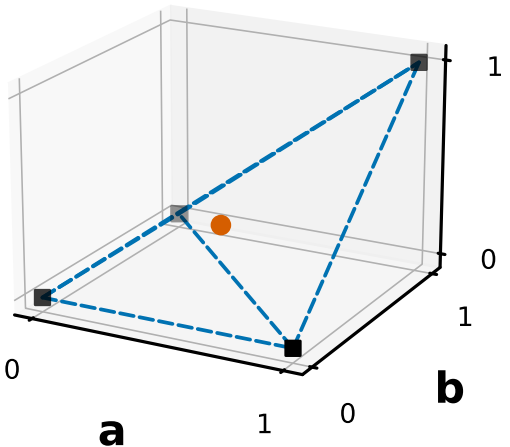

$$\max(0, a+b-1) \le q \le \min(a, b)$$

Weights: $(1-a-b+q,\ a-q,\ b-q,\ q)$

Corners: (0,0,0), (1,0,0),
(0,1,0), (1,1,1)

The marked point is (.4, .6, .2).